# A decision rule without an operating characteristic: the pairwise sum in conformance assessment of multi-channel boresight alignment

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**KEYWORDS**

conformity assessment; decision rule; measurement uncertainty; boresight alignment; tolerance allocation; operating characteristic; minimal enclosing circle; guard band

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**DECLARATION OF COMPETING INTEREST**

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

**CREDIT AUTHOR STATEMENT**

Zhengkang Zuo: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization.

**ETHICS**

Not applicable. This study did not involve human participants, human data or animal subjects.

**DATA AVAILABILITY**

The allocation table transcribed from the public specification, the solver, the sweep drivers, the derived CSV outputs and the figure scripts are provided as supplementary material with this submission. On acceptance the same package will be deposited in a public repository and the DOI added to the article. The specification itself is a public NASA document and is not redistributed.

**DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES**

During the preparation of this work the author used a large language model assistant to write code and to polish the language of the manuscript. The author reviewed and edited all content, verified every numerical result against the supplied code and data, and takes full responsibility for the content of the publication.